ALICE L WALTON SCHOOL OF MEDICINE

Genetic Thrombotic Disorders

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Adapted by: Chelsey Deel

Editor: Jordan Finn

Adapted from: Genetic Thrombotic Disorders

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0:00 / 26:25 1x

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of genetic thrombotic disorders.
2. Explain the pathophysiology of genetic thrombotic disorders.
3. Interpret the clinical findings to properly diagnose genetic thrombotic disorders.
4. Outline the management plan of genetic thrombotic disorders.

CASE CONNECTION

"I don't smoke. I'm not old. I'm not on birth control pills. I'm not inactive. I haven't had surgery. I'm not overweight. And I'm certainly not pregnant. When I did some online research, those are all listed as the risk factors for blood clots. So, I don't understand why I got another one. What should I do?" KJ asks. She is a 42-year-old female referred for evaluation of a recurrent deep vein thrombosis (DVT) and one episode of a pulmonary embolism. She has no relevant medical history and takes no medications. Her physical examination is normal.

How will you respond to KJ's question? What workup should be pursued? Consider your answers as you read, and we'll revisit KJ at the end of the brick.

GO TO CONCLUSION ↓

What Are Genetic Thrombotic Disorders?

Abnormal blood clots are a common problem, usually associated with risk factors such as smoking, cancer, increased estrogen levels, immobilization, or recent major surgery. Sometimes, however, patients with none of these risk factors develop thrombi. Some have an underlying thrombotic disorder that is responsible for their thrombus formation. These can be hereditary or acquired. Here, we'll address the uncommon hereditary thrombotic disorders.

The overall incidence of the associated gene mutations is 8% to 10% in the general population, b[illegible] **velop thrombi**. Among those who do, the disorders can account for up to 40% of cases, with the highest frequency in White populations. These genetic disorders should be tested for when patients have thrombi without other risk factors.

How Do Patients With Genetic Thrombotic Disorders Present (LO #1)?

Not all venous thrombi have symptoms, and many resolve spontaneously. When they present with symptoms, thrombi usually occur in the venous (not the arterial) system. The most common site is the deep veins of the leg (**deep vein thrombosis** [**DVT**]), and patients present with unilateral leg/calf swelling and tenderness. The patient can also present with an embolus that travels to a distant site from the DVT, which is usually a **pulmonary embolus** that can cause dyspnea and chest pain.

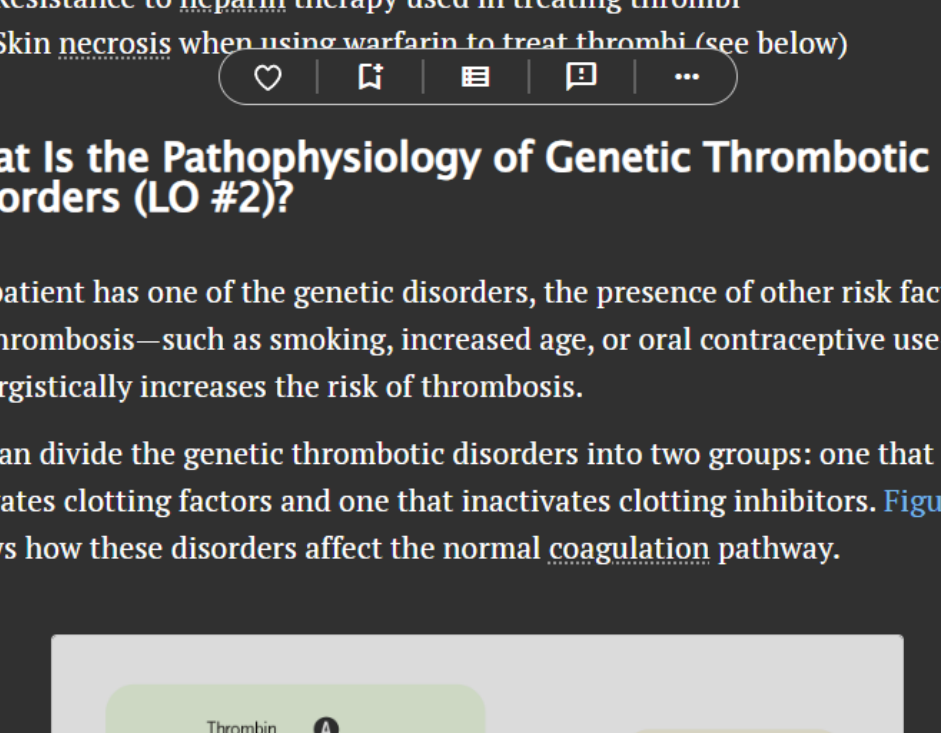
If you are evaluating a patient with a DVT or other thrombotic disorder, certain factors suggest a genetic cause:

- Lack of established risk factors for thrombi, such as smoking, cancer, increased estrogen levels (e.g., pregnancy, oral contraceptives, hormone replacement therapy), immobilization, and recent major surgery
- Thrombi in a young patient (<45 years)
- Family history of thrombosis
- Personal history of previous thrombi
- Presence of thrombi in unusual sites, such as the cerebral, mesenteric, or portal veins
- Resistance to heparin therapy used in treating thrombi
- Skin necrosis when using warfarin to treat thrombi (see below)

What Is the Pathophysiology of Genetic Thrombotic Disorders (LO #2)?

If a patient has one of the genetic disorders, the presence of other risk factors for thrombosis—such as smoking, increased age, or oral contraceptive use—synergistically increases the risk of thrombosis.

We can divide the genetic thrombotic disorders into two groups: one that activates clotting factors and one that inactivates clotting inhibitors. Figure 1 shows how these disorders affect the normal coagulation pathway.



Using Figure 1 as a guide, we will explain the mechanism for increased thrombosis in each of the main disorders.

Disorders That Activate Thrombosis

This category includes the second most common genetic thrombotic disorder: factor II gene mutation.

Factor II (Prothrombin G20210A) Gene Mutation. These patients have a gene mutation that leads to overproduction of factor II, stimulating thrombosis (see 2 in Figure 1). Most patients have a point mutation in the gene for factor II that results in a substitution of guanine for adenine at position 20210 of the prothrombin gene. More prothrombin means more thrombin, which leads to more conversion of fibrinogen to fibrin. This results in an increased tendency to form blood clots. It is more common in patients of Northern European descent and is inherited in an autosomal dominant pattern with variable penetrance.

Disorders That Block Thrombosis Termination

Unlike factor II mutation, these disorders do not accelerate thrombosis. Instead, they prevent thrombosis from being terminated or regulated. This involves abnormal function of two main inhibitors of coagulation, protein C or S, and antithrombin (see Figure 1).

Factor V Leiden. Factor V Leiden is the most common hereditary thrombotic disease, accounting for 40% to 50% of all cases of unexplained thrombosis. Patients with a single factor V Leiden gene mutation have an approximately fivefold increased ris[illegible]mozygous factor V Leiden gene mutations have a much higher risk. Named after the city in the Netherlands where the mutation was discovered, factor V Leiden has since been found in people worldwide (although it is much more common in the White population, particularly those of European ancestry).

Factor V Leiden is an abnormal factor V molecule made from a hereditary **mutated factor V gene**. Normal factor V is activated by thrombin to become factor Va, which is the active form of factor V (see 5 in Figure 1). In this disorder, the factor V Leiden gene contains a single-point mutation that results in an arginine-to-glutamine substitution in the factor V protein. This mutation happens precisely at the location where protein C usually cleaves (and inactivates) factor V. What this means is that factor V Leiden can participate in the coagulation cascade just like regular factor V can. But, when the time comes to stop making fibrin, protein C cannot cleave (and inactivate) factor V Leiden. Thus, thrombosis increases abnormally, leading to clotting.

What is the mechanism of factor V Leiden?

Protein C or S Deficiency. Protein C, when it becomes activated, terminates coagulation by directly inactivating factors Va and VIIIa (see Figure 1). The normal function of protein S is to act as a cofactor to enable the function of protein C. This means that without protein S, protein C cannot do its job. Without the joint anticoagulant effects of proteins C and S, the blood is more prone to clotting.

Deficiencies in proteins C and S can be inherited or acquired. Acquired deficiencies most typically occur with liver disease, vitamin K deficiency, and some chemotherapy agents. Inherited deficiencies are caused by genetic mutations in the gene for either protein C or S. Patients who are homozygous in either deficiency are much more prone to problematic clotting.

Complications can occur. **Warfarin-induced skin necrosis** is a potentially lethal condition seen in patients with protein C deficiency. It causes skin and subcutaneous tissue thrombosis and necrosis. The anticoagulant **warfarin** blocks the production of vitamin K–dependent coagulation factors, which include proteins C and S, as well as factors II, VII, IX, and X. Of all these vitamin K–dependent coagulation factors, proteins C and factor VII have the shortest half-life. So, after warfarin administration, the first proteins to decrease are protein C and factor VII. The effects on protein C tend to predominate, and this puts the patient in a transient hypercoagulable state, even though they have just taken an anticoagulant medication.

This state is normally transient and does not cause problems. However, if you give warfarin to a patient who has protein C deficiency, they may develop skin thrombosis, with symptoms ranging from painful, erythematous lesions to bulla formation to ou[illegible]2).

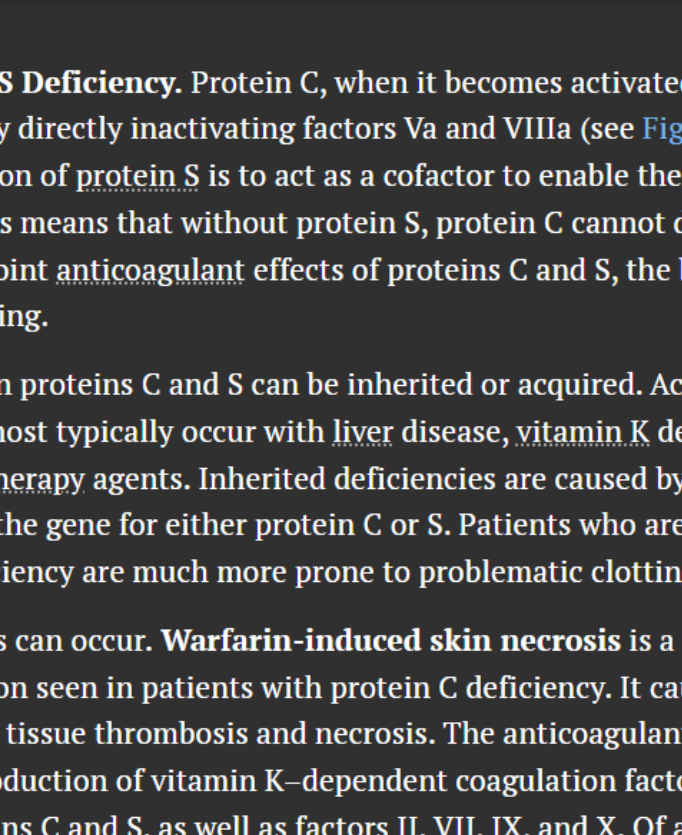


Figure 2

This effect is limited to protein C deficiency; protein S deficiency rarely leads to warfarin-induced skin necrosis.

CLINICAL CORRELATION

Patients with protein C[illegible]**fulminans**, an acute disorder characteriz[illegible]atients with purpura fulminans may present with sudden-onset hemorrhagic necrosis and infarction of the skin, caused by widespread, multiple thrombi. Patients with purpura fulminans frequently have associated **disseminated intravascular coagulation**, often with fatal consequences.

Why does initiating treatment with warfarin induce a period of hypercoagulability?

Antithrombin Deficiency. Now let's look at an abnormality in the other main regulatory pathway of thrombosis. Antithrombin (AT) is an enzyme inhibitor that inhibits coagulation by inhibiting thrombin (factor IIa) as well as activated factors VIIa, IXa, Xa, XIa, and XIIa. So, individuals with AT deficiency are prone to developing thrombotic episodes.

Although AT deficiencies may be inherited or acquired, we are mainly focusing on the hereditary for[illegible]**vels of AT or defective AT**. The pattern of inheritance is autosomal dominant and is caused by mutations in the antithrombin gene. Unlike factor V Leiden and factor II mutations, hereditary AT deficiency is not defined by a single specific mutation. Rather, various types of mutations at different locations in the antithrombin gene may lead to this disorder.

An interesting complication of this disorder is **resistance to the anticoagulant drug heparin**, which is used to treat many thrombotic disorders.

CLINICAL CORRELATION

This is uncommon and is mostly seen in patients who are severely ill and on cardiopulmonary bypass or extracorporeal membrane oxygenation (ECMO)—generally, patients with heart failure in the intensive care unit.

Why does this happen? Heparin works by binding to antithrombin and inducing a conformational change that increases AT affinity for thrombin and factor X. This dramatically increases the inactivating ability of antithrombin and promotes anticoagulation. In AT deficiency, reduced antithrombin activity would naturally lead to a reduced effect for heparin. Patients with AT deficiency often need higher doses of heparin to achieve a proper level of anticoagulation. Patients who need anticoagulation can be supplemented with antithrombin or use another anticoagulant that is not dependent on antithrombin, such as dabigatran.

Why do patients with antithrombin deficiency exhibit heparin resistance?

How Do We Diagnose Genetic Thrombotic Disorders (LO #3)?

Individuals with recurrent, unprovoked, or unusual thrombosis at a young age; family history of thrombosis; heparin resistance; or warfarin-induced skin necrosis should be considered for further diagnostic workup of these disorders. This may include:

- Detect factor V Leiden by genetic testing or activated protein C resistance assays
- Measure protein S or C levels or activity
- Test for factor II gene mutation by genetic analysis
- Measure plasma antithrombin activity (AT-heparin cofactor assay)

Which testing should be performed if factor V Leiden is suspected in a patient?

How Do We Treat Genetic Thrombotic Disorders (LO #4)?

Because most of these patients present with an unexplained thrombus, they are treated with an appropriate anticoagulant such as heparin, warfarin, or direct oral anticoagulants (e.g., dabigatran, apixaban).

Patients with protein C or S deficiency should use warfarin with caution because of the risk of skin necrosis.

Preventing Thrombi

Whether to use prophylactic anticoagulation to prevent DVT in patients with these disorders is a complex decision. A few generalizations are the following:

- Anticoagulation [illegible]eded.
- Select patients are given preventive prophylaxis with anticoagulants before surgery.
- Pregnant patients with a history of thrombi and some high-risk patients are given anticoagulants both before and after delivery.

In general, the majority of patients with genetic thrombophilia will never develop thrombosis; therefore, most patients do not require long-term prophylactic anticoagulation.

Which genetic thrombotic disorder should not be treated with warfarin?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to KJ, how do you respond to her question? What workup should be pursued?

"The information yo[illegible]hings do increase the risk for a blood clo[illegible]you have a gene mutation that can't be turned off. This mutation causes the formation of a protein involved in the formation of blood clots. In essence, the engine for clot formation is turned on, but you can't remove the key from the ignition to turn it off when you need to."

Bloodwork is ordered, and KJ is positive for the factor V Leiden mutation.

Summary

- Genetic thrombotic disorders predispose patients to increased clotting and thrombus formation. These disorders may be seen in up to 40% of patients with thrombi.
- Thrombi usually occur in the venous (not the arterial) system. Patients are at risk for deep vein thrombosis and pulmonary embolus.
- Factor V Leiden is the most common hereditary thrombotic disorder. The underlying defect in factor V Leiden is a point mutation in the factor V gene, resulting in a factor V Leiden molecule that cannot be inactivated by protein C.
- Patients presenting with thrombi in unusual locations (such as in the mesenteric or cerebral circulation) or young patients without classic risk factors for thrombosis should raise suspicion for underlying genetic thrombotic disorders.
- Factor II genetic mutations lead to high levels of factor II.
- Protein C or S deficiencies lead to a lack of inactivation of the clotting cascade, with increased thrombotic risk. Warfarin-induced skin necrosis is a complication of protein C deficiency.
- Hereditary antith[illegible]genetic mutations in the antithrombin gene that lead to impaired antithrombin inactivation of clotting.
- Diagnosis is with genetic tests for factor V Leiden and factor II, serum levels of protein C or S, and a functional assay for antithrombin.
- Patients with these disorders and thrombi should be treated with anticoagulants.
- Prophylaxis with anticoagulants before surgery or during pregnancy should be considered for some patients.

Review Questions

1. In patients with a factor V Leiden (FVL) mutation, what is wrong with the protein?

A. ● FVL cannot be inactivated by protein C.

B. ○ FVL cannot interact with factor X.

C. ○ FVL is immediately cleaved to its inactive form.

D. ○ FVL is present at suboptimal levels.

E. ○ FVL shows increased reactivity to prothrombin.

Hide Explanation

The correct answer is[illegible]. People with factor V Leiden (FVL) have a mutation in the gene for factor V. Factor V Leiden is an abnormal version of factor V that is resistant to the action of activated protein C (APC). So, APC cannot easily stop factor V Leiden from making more fibrin. Once the coagulation process is turned on in people with factor V Leiden, it turns off more slowly than in people with normal factor V. FVL cannot interact with factor X (B) is incorrect; it does so just like normal factor V. FVL is immediately cleaved to its inactive form (C) is incorrect; in fact, the whole problem is that FVL can't be cleaved to its inactive form. FVL is present at suboptimal levels (D) is incorrect; it is present in normal levels in FVL disease. FVL shows increased reactivity to prothrombin (E) is incorrect; FVL does not show any change in reactivity to prothrombin.

2. A 23-year-old male presents to the emergency department with his third episode of unilateral lower-extremity swelling and pain. He is diagnosed with a deep vein thrombosis and worked up for thrombophilia. His tests come back positive for a mutation in the gene encoding factor II. How does factor II genetic mutation lead to hypercoagulability?

A. ○ Decreased amount of antithrombin

B. ○ Increased activation of prothrombin

C. ○ Increased activity of thrombin

D. ● Increased production of prothrombin

E. ○ Resistance to antithrombin

Hide Explanation

The correct answer is increased production of prothrombin (D). Factor II genetic mutation leads to hypercoagulability by increasing the amount of prothrombin produced. The increased amount of prothrombin leads to more prothrombin converted to thrombin, which converts fibrinogen to fibrin and causes clot formation. Antithrombin deficiency causes decreased amount of antithrombin (A). This would lead to more thrombin activity but through a different mechanism. Increased activation of prothrombin (B), increased activity of thrombin (C), and resistance to antithrombin (E) is seen in other conditions but is not commonly seen with factor II.

3. A 44-year-old female presents to the office for a follow-up visit. She has had several episodes of deep vein thrombosis (DVT) and was ultimately found to have a deficiency in protein C. Which of the following mechanisms causes this patient's recurrent DVTs?

A. ○ Factor Va resistance to deactivation

B. ○ Increased blood stasis

C. ● Increased factor Va and VIIIa activity

D. ○ Increased prothrombin production

E. ○ Loss of cofactor activity

Hide Explanation

The correct answer is increased factor Va and VIIIa activity (C). Protein C normally inactivates those two factors in an effort to safely control clotting. With less protein C, there is more activity from factors Va and VIIIa. Factor Va resistance to deactivation (A) is incorrect. A protein Va that is resistant to deactivation is seen in factor V Leiden. This is similar to increased activity of factors Va and VIIIa, but factor V Leiden only has increased factor Va activity. Increased blood stasis (B) is one method of producing a provoked deep vein thrombosis, such as in prolonged immobility. These often occur after a long plane ride or in long-distance truckers. Increased prothrombin production (D) is the mechanism by which prothrombin gene mutation induces hypercoagulability. Loss of cofactor activity (E) is why protein S deficiency produces coagulability. Protein S normally functions as a cofactor for protein C. Without the cofactor activity of protein S, protein C cannot work.

4. A 22-year-old male presents to the emergency department with left lower-leg swelling and pain. He has had multiple episodes with this presentation. Which of the following underlying disorders is most likely responsible for these recurrent symptoms?

A. ○ Antiphospholipid syndrome

B. ● Factor V Leiden

C. ○ Malignancy

D. ○ Protein C deficiency

E. ○ Prothrom[illegible]

Hide Explanation

The correct answer is factor V Leiden (B). This patient presents with recurrent blood clots in his lower extremities that suggest he has a genetic disorder predisposing him to developing thrombi. Factor V Leiden is the most common hereditary thrombotic condition and therefore is the best answer. Antiphospholipid syndrome (A) may cause blood clots such as DVT but is also associated with multiple miscarriages or stillbirths in female patients. Malignancy (C) creates a hypercoagulable state, but nothing suggests that this young, healthy patient has cancer. Protein C deficiency (D) is less common than is factor V Leiden. Prothrombin gene mutation (E) also results in hypercoagulability but is less common than is factor V Leiden.

5. A 25-year-old male comes to the ER complaining of sudden onset of redness and swelling of his right lower extremity. His father had a pulmonary embolus at age 50; his 33-year-old sister had a blood clot in her leg during pregnancy. Compression Doppler ultrasound of his right leg demonstrates a thrombus in his right femoral vein, consistent with a DVT. Further testing confirms he is heterozygous for the Factor V Leiden mutation. What is the initial treatment for the patient's DVT?

A. ○ Aspirin, elevation, and warm soaks to the leg

B. ○ Low-molecular weight heparin only

C. ● Low-mol[illegible]

D. ○ Oral warfarin only

Hide Explanation

The correct answer is LMW heparin and warfarin (C). Aspirin, elevation, and warm soaks to the leg (A) are not adequate therapy for a femoral vein DVT. LMW heparin only (B) is incorrect because heparin is administered parenterally; as the patient is going to require several months of anticoagulation, an oral option such as warfarin is preferred. Oral warfarin only (D) is incorrect because it takes several days for warfarin to attain therapeutic levels (patients can develop worsening thrombosis or embolism in that time) and starting warfarin by itself can result in skin necrosis. Therefore, the common treatment is using LMW heparin and warfarin together, with heparin acting as a bridge until warfarin becomes therapeutic for 2 days. The fact that the patient has an inherited thrombophilia does not affect the initial treatment of the DVT.

How would you rate this Brick?

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Differentiate the clinical presentation of genetic thrombotic disorders.~~

02 ☑ ~~Explain the pathophysiology of genetic thrombotic disorders.~~

03 ☑ ~~Interpret~~[illegible]~~genetic thromb~~

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- https://www.uptodate.com/contents/factor-v-leiden-and-activated-protein-c-resistance?search=genetic%20thrombotic%20disorders&source=search_result&selectedTitle=5~150&usage_type=default&display_rank=5
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Objective Confidence: 4/4